

Kevin Pulikkottil

San Jose, CA | 669-253-8966 | kevinpul101@gmail.com

<https://www.linkedin.com/in/kpulik> | <https://github.com/kpulik> | <https://kpulik.github.io/Portfolio>

SUMMARY

Computer Science graduate with experience in software development, systems design, and network security, seeking entry-level Software or Cybersecurity Engineer roles.

EDUCATION

The University of Texas at Dallas, Richardson, TX

May 2026

Bachelor of Science, Computer Science

- Coursework: Database Systems, Computer Networks, Networking Laboratory, Operating Systems, Data Structures & Algorithm Analysis, Software Engineering, Digital Logic & Computer Design

EXPERIENCE

Toyota Secure Vehicle Communications, UTDesign Capstone Project

January 2026 – May 2026

Vehicle Cybersecurity Engineer, Embedded Security Project

Tools Used: C/C++, Python, Arm64 QEMU, TEE, X448

- Built an **end-to-end, X448-based, rekey-capable EV charging handshake** in the **OP-TEE Trusted Application** and host client, meeting Toyota TMNA Product Cybersecurity standards for the **ISO 15118-20** use case.
- Implemented **X448 Elliptic Curve Diffie-Hellman (RFC 8418)** for ephemeral session keys and stabilized **X.509 certificates** across sessions by refactoring the custom CMAC extension out of the TA, builder, parser, and verifier.
- Replaced the team's PyQt6 desktop demo with a browser-based dashboard for the UTDesign EXPO, visualizing **live MITM attack detection** between EV, EVSE, and a Raspberry Pi attacker node for corporate stakeholders.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, SQL

Software Engineering: Object-Oriented Design, Data Structures, Algorithms, Software Design Patterns, UNIX Systems Programming, Database Design and Normalization

Tools & Platforms: Git, Linux, Arm64 QEMU, Raspberry Pi

Security: Network Security, Firewalls, Network Protocols (OSI Model, TCP/IP), Access Control, Data Integrity

PROJECTS & OPEN SOURCE

MemPalace, Open Source Contributor (GitHub)

April 2026

Tools Used: Python, CLI, MCP, Pytest

- Contributed a merged fix to MemPalace (**55,000+ GitHub stars**) resolving a user-reported CLI bug, adding a first-class *mempalace mcp* subcommand that prints copy-pastable MCP server setup guidance with custom palace-path handling, backed by regression tests.

Secure Home Lab on Raspberry Pi, Personal

December 2025 – Present

Tools Used: Raspberry Pi, Linux, Tailscale, Pi-hole, DNS Configuration, Firewall Management

- Reduced network ad traffic by **99%** by deploying a Pi-hole DNS sinkhole reachable remotely over a Tailscale tailnet.
- Implemented custom firewall rules, strict access controls, and active DNS query log monitoring.

Breast Cancer Detection Model, Algoverse AI Research

February 2024 – May 2024

Tools Used: Python, scikit-learn, pandas, NumPy

- **Achieved 95.6% diagnostic accuracy** by building and tuning a scikit-learn classifier on the Breast Cancer Wisconsin dataset.

CERTIFICATIONS & AWARDS

Eagle Scout, Scouting America

January 2021

CompTIA Security+, CompTIA

April 2026

CDSA (Certified Defensive Security Analyst), HackTheBox

In Progress